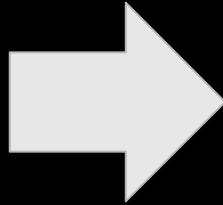


# Certified Adversarial Robustness Via Randomized Smoothing

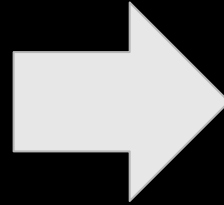
# Background



**Classifier**



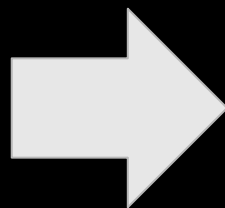
**Classifier**



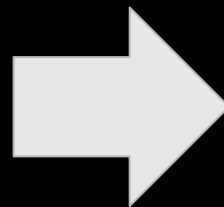
U

U

U



**Classifier**



**Monkey**

**System that receives an input and  
outputs a categorized output**



# **Adversarial Perturbation**



U

U



U

U

U

**Small changes added to the image to  
fool the classifier**

u

u

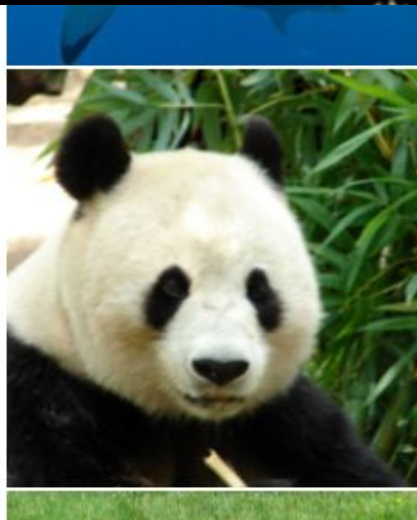
$\delta$  = Adversarial Perturbation

u

$$x' = x + \delta$$



**Gaussian Noise**



**Random noise added to the image to  
test stability in the classifier**

$$x' = x + \epsilon$$

$$\epsilon \sim N(0, \sigma^2 I)$$



**Certified  
Robustness**

**A guaranteed radius around an input  
which will not change the classifier's  
output**

# Research Motivation

- **Classifiers are vulnerable to adversarial perturbations**
- **Prior randomized smoothing had loose guarantees**
- **Scale poorly to large datasets like ImageNet**

# Research Problem

**Create a classifier utilizing the technique of randomized smoothing to provide a tight robustness guarantee on large datasets**

# Innovation

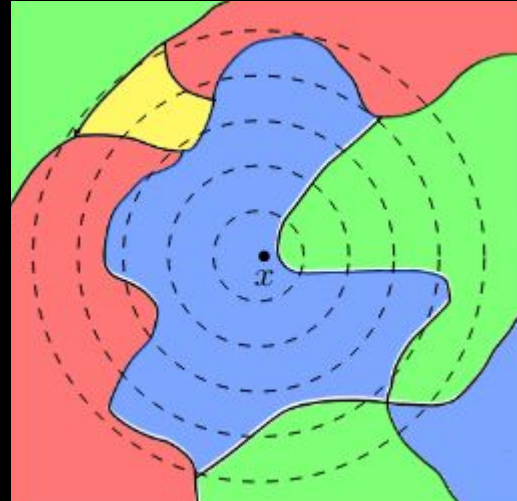
# Randomized Smoothing

What makes in random? What makes in smoothing?

The picture is  
randomizing the  
Gaussian noise in the  
image and smoothing  
is taking into account  
many noisy inputs



$$g(x) = \operatorname{argmax} P(f(x+\epsilon) = c)$$



$$P(f(x+\epsilon)=c)$$

**What is the probability of class  $c$  from the classified image with Gaussian noise**

**$\sigma$  controls standard deviation of the Gaussian noise**

**Larger  $\sigma$  means a bigger Gaussian cloud and vice versa**

## **Relationship, accuracy-robustness tradeoff**

**Larger  $\sigma$  more smoothing -> greater robustness potent.**

**But more corruption to image and worse accuracy potent.**

$$R = \sigma/2(\Phi^{-1}(p_A) - \Phi^{-1}(p_B))$$

**Theorem 1 is saying it can find a radius  
and Theorem 2 is saying that it is tight**

## Theorem 1

Let's have Gaussian samples around  $x$  produce the top class  $c_A$ , thus  $p_A$  is the probability of the winning class.

And  $p_B$  be the probability of the second highest

$$\mathbb{P}(f(x + \varepsilon) = c_A) \geq \underline{p_A} \geq \overline{p_B} \geq \max_{c \neq c_A} \mathbb{P}(f(x + \varepsilon) = c) \quad (2)$$

Most important to note that  $p_A$  is always greater or at least equal to the probability of  $p_B$

*Then  $g(x + \delta) = c_A$  for all  $\|\delta\|_2 < R$ , where*

$$R = \frac{\sigma}{2}(\Phi^{-1}(\underline{p}_A) - \Phi^{-1}(\overline{p}_B)) \quad (3)$$

## Certified robustness guarantee

**R = certified radius,  $\Phi^{-1}$  = reverse of CDF of the standard normal distribution**

*Then  $g(x + \delta) = c_A$  for all  $\|\delta\|_2 < R$ , where*

$$R = \frac{\sigma}{2}(\Phi^{-1}(\underline{p}_A) - \Phi^{-1}(\overline{p}_B)) \quad (3)$$

**Find the gaussian distance and  
multiply it by the Gaussian noise**

*Then  $g(x + \delta) = c_A$  for all  $\|\delta\|_2 < R$ , where*

$$R = \frac{\sigma}{2}(\Phi^{-1}(\underline{p}_A) - \Phi^{-1}(\overline{p}_B)) \quad (3)$$

**Relationships: Can anyone tell me?**

## Theorem 2

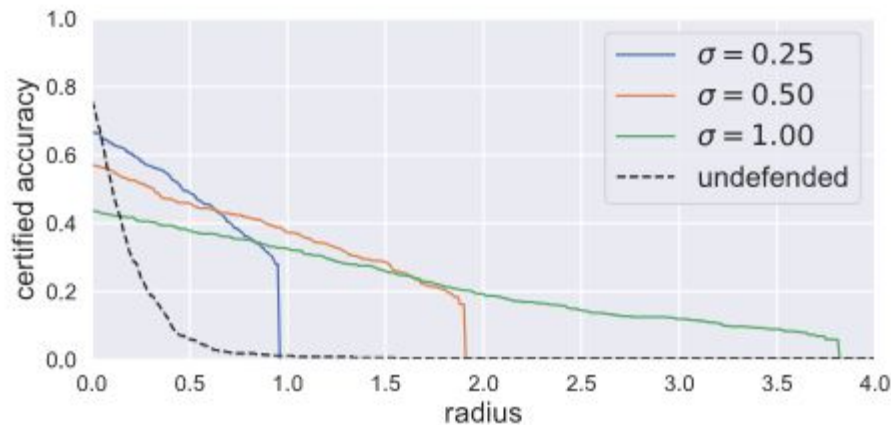
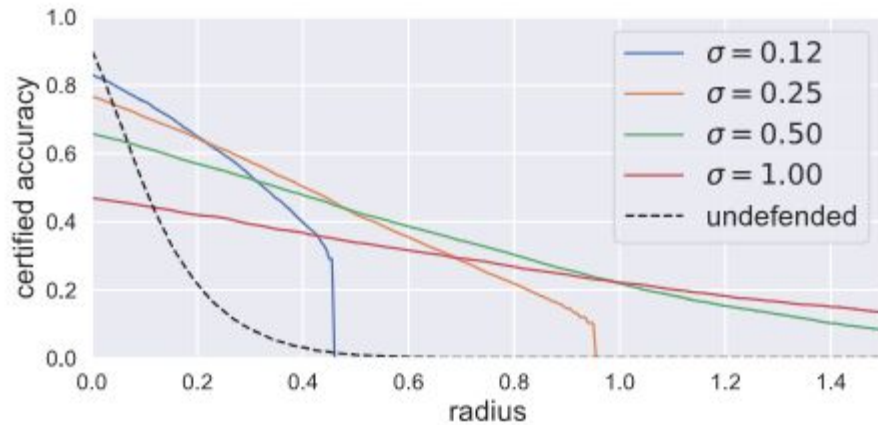
Given only the probability of class A and B, the certified radius is tight

# Practical Algorithms

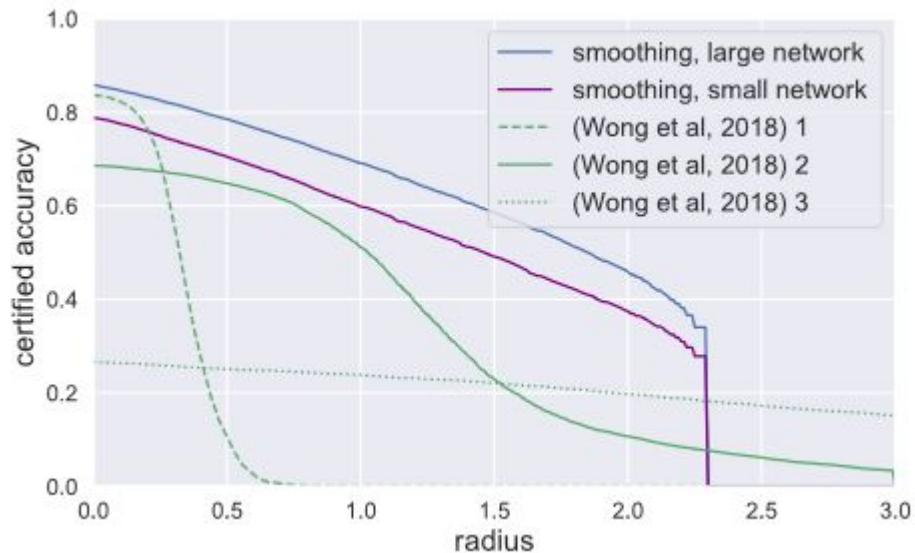
**PREDICT:** Add random Gaussian noise many times and choose the class that appears the most

**CERTIFY:** Use noisy samples to calculate how far the input can be changed without changing prediction

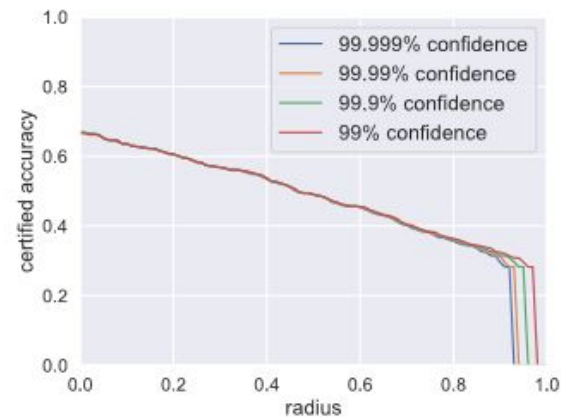
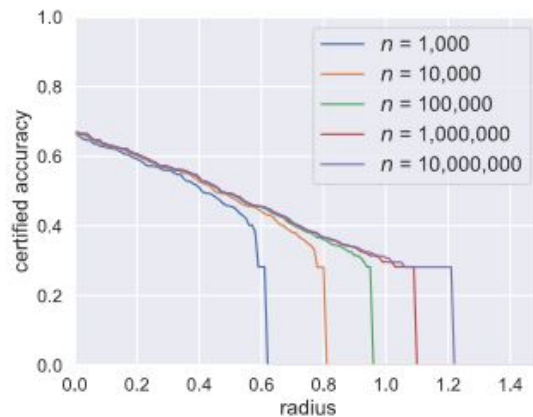
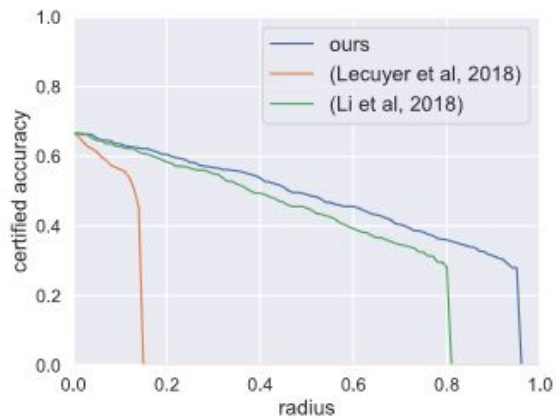
# Evaluation Check



**Shows accuracy  
-robustness tradeoff,  
meaning the larger the  
certified radius gets the  
less accurate it becomes**



**These graphs shows the difference between the certified-robustness method compared to the smoothing method**



These graphs shows the difference between the certified-robustness method compared to the smoothing method

# Future Work



- Exploring other noise distributions to certify robustness in something other than L2
- Develop better ways to train classifiers to remain accurate under noise
- Improve the computational efficiency of randomized certification

Thank you